

8.61 Find the value of the resistance R_e , which, when connected in the emitter lead of a CE BJT amplifier, raises the output resistance by a factor of (a) 5, (b) 10, and (c) 50. What is the maximum possible factor by which the output resistance can be raised, and at what value of R_e is it achieved? Assume the BJT has $\beta = 100$ and is biased at $I_C = 0.5$ mA.

Section 8.5: The Cascode Amplifier

D 8.62 In a MOS cascode amplifier, the cascode transistor is required to raise the output resistance by a factor of 50. If the transistor is operated at $V_{OV} = 0.2$ V, what must its V_A be? If the process technology specifies V_A' as 5 V/ μm , what channel length must the transistor have?

D 8.63 For a cascode current source such as that in Fig. 8.32, show that if the two transistors are identical, the current I supplied by the current source and the output resistance R_o are related by $IR_o = 2|V_A'|^2/|V_{OV}|$. Now consider the case of transistors that have $|V_A| = 4$ V and are operated at $|V_{OV}|$ of 0.2 V. Also, let $\mu_p C_{ox} = 100$ $\mu\text{A}/\text{V}^2$. Find the W/L ratios required and the output resistance realized for the two cases: (a) $I = 0.1$ mA and (b) $I = 0.5$ mA. Assume that V_{SD} for the two devices is the minimum required (i.e., $|V_{OV}|$).

D*8.64 For a cascode current source, such as that in Fig. 8.32, show that if the two transistors are identical, the current I supplied by the current source and the output resistance R_o are related by

$$IR_o = \frac{2|V_A'|^2}{|V_{OV}|} L^2$$

Now consider the case of a 0.18- μm technology for which $|V_A'| = 5$ V/ μm and let the transistors be operated at $|V_{OV}| = 0.2$ V. Find the figure-of-merit IR_o for the three cases of L equal to the minimum channel length, twice the minimum, and three times the minimum. Complete the entries of the table at the bottom of the page. Give W/L and the area $2WL$ in terms of n , where n is the value of W/L for the case $I = 0.01$ mA. In the table, A_v denotes the gain obtained in a cascode amplifier such as that in Fig. 8.33 that utilizes our current source as load and which has the same values of g_m and R_o as the current-source transistors.

- For each current value, what is price paid for the increase in R_o and A_v obtained as L is increased?
- For each value of L , what advantage is obtained as I is increased, and what is the price paid? (*Hint:* We will see in Chapter 10 that the amplifier bandwidth increases with g_m .)
- Contrast the performance obtained from the circuit with the largest area with that obtained from the circuit with the smallest area.

D 8.65 Design the cascode amplifier of Fig. 8.30(a) to obtain $g_{m1} = 2$ mA/V and $R_o = 200$ k Ω . Use a 0.18- μm technology for which $V_{tn} = 0.5$ V, $V_A' = 5$ V/ μm , and $k_n' = 400$ $\mu\text{A}/\text{V}^2$. Determine L , W/L , V_{G2} , and I . Use identical transistors operated at $V_{OV} = 0.25$ V, and design for the maximum possible negative signal swing at the output. What is the value of the minimum permitted output voltage?

8.66 The cascode amplifier of Fig. 8.33 is operated at a current of 0.2 mA with all devices operating at $|V_{OV}| = 0.20$ V.

	$L = L_{\min} = 0.18 \mu\text{m}$ $IR_o = \quad \text{V}$				$L = 2L_{\min} = 0.36 \mu\text{m}$ $IR_o = \quad \text{V}$				$L = 3L_{\min} = 0.54 \mu\text{m}$ $IR_o = \quad \text{V}$			
	g_m (mA/V)	R_o (k Ω)	A_v (V/V)	$2WL$ (μm^2)	g_m (mA/V)	R_o (k Ω)	A_v (V/V)	$2WL$ (μm^2)	g_m (mA/V)	R_o (k Ω)	A_v (V/V)	$2WL$ (μm^2)
$I = 0.01$ mA $W/L = n$												
$I = 0.1$ mA $W/L =$												
$I = 1.0$ mA $W/L =$												

All devices have $|V_A| = 4$ V. Find g_{m1} , the output resistance of the amplifier, R_{on} , the output resistance of the current source, R_{op} , the overall output resistance, R_o , and the voltage gain, A_v .

D 8.67 Design the CMOS cascode amplifier in Fig. 8.33 for the following specifications: $g_{m1} = 1$ mA/V and $A_v = -280$ V/V. Assume that for the available fabrication process, $|V'_A| = 5$ V/ μm for both NMOS and PMOS devices and that $\mu_n C_{ox} = 4 \mu_p C_{ox} = 400 \mu\text{A/V}^2$. Use the same channel length L for all devices and operate all four devices at $|V_{OV}| = 0.25$ V. Determine the required channel length L , the bias current I , and the W/L ratio for each of four transistors. Assume that suitable bias voltages have been chosen, and neglect the Early effect in determining the W/L ratios.

D 8.68 Design the circuit of Fig. 8.32 to provide an output current of $100 \mu\text{A}$. Use $V_{DD} = 3.3$ V, and assume the PMOS transistors to have $\mu_p C_{ox} = 60 \mu\text{A/V}^2$, $V_{tp} = -0.8$ V, and $|V_A| = 5$ V. The current source is to have the widest possible signal swing at its output. Design for $V_{OV} = 0.2$ V, and specify the values of the transistor W/L ratios and of V_{G3} and V_{G4} . What is the highest allowable voltage at the output? What is the value of R_o ?

8.69 The cascode transistor can be thought of as providing a “shield” for the input transistor from the voltage variations at the output. To quantify this “shielding” property of the cascode, consider the situation in Fig. P8.69. Here we have grounded the input terminal (i.e., reduced v_i to zero), applied a small change v_x to the output node, and denoted the voltage change that results at the drain of Q_1 by v_y . By what factor is v_y smaller than v_x ?

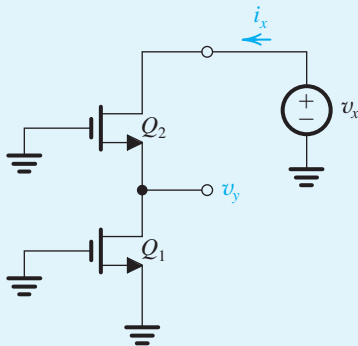


Figure P8.69

***8.70** In this problem we investigate whether, as an alternative to cascoding, we can simply increase the channel length L of the CS MOSFET. Specifically, we wish to compare the

two circuits shown in Fig. P8.70(b) and (c). The circuit in Fig. P8.70(b) is a CS amplifier in which the channel length has been quadrupled relative to that of the original CS amplifier in Fig. P8.70(a) while the drain bias current has been kept constant.

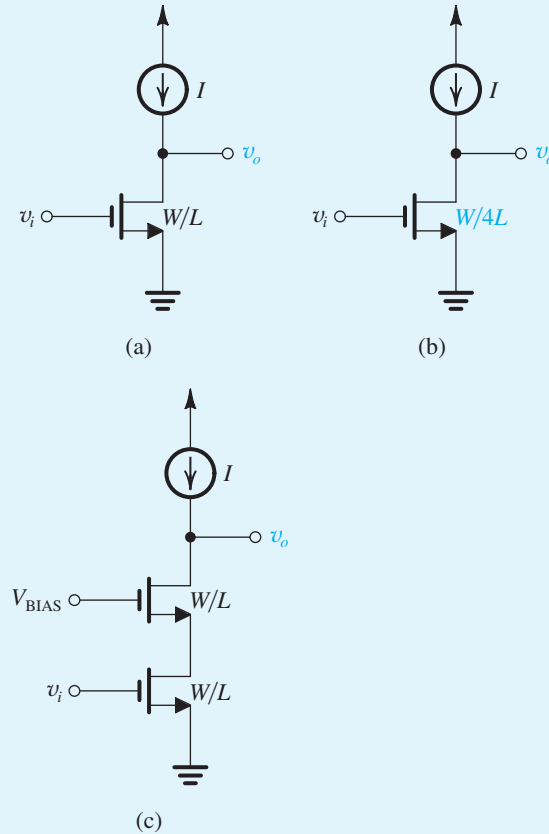


Figure P8.70

- Show that for this circuit V_{OV} is double that of the original circuit, g_m is half that of the original circuit, and $\frac{v_o}{v_i}$ is double that of the original circuit.
- Compare these values to those of the cascode circuit in Fig. P8.70(c), which is operating at the same bias current and has the same minimum voltage requirement at the drain as in the circuit of Fig. P8.70(b).

8.71 Consider the cascode amplifier of Fig. 8.33 with the dc component at the input $V_i = 0.6$ V, $V_{G2} = 0.9$ V, $V_{G3} = 0.4$ V, $V_{G4} = 0.7$ V, and $V_{DD} = 1.3$ V. If all devices are matched, that is, $k_{n1} = k_{n2} = k_{p3} = k_{p4}$, and have equal $|V_t|$ of 0.4 V, what is the overdrive voltage at which the four transistors are operating? What is the allowable voltage range at the output?

SIM 8.72 A CMOS cascode amplifier such as that in Fig. 8.34(a) has identical CS and CG transistors that have $W/L = 5.4 \mu\text{m}/0.36 \mu\text{m}$ and biased at $I = 0.2 \text{ mA}$. The fabrication process has $\mu_n C_{ox} = 400 \mu\text{A}/\text{V}^2$, and $V_A' = 5 \text{ V}/\mu\text{m}$. At what value of R_L does the gain become -100 V/V ? What is the voltage gain of the common-source stage?

8.73 The purpose of this problem is to investigate the signal currents and voltages at various points throughout a cascode

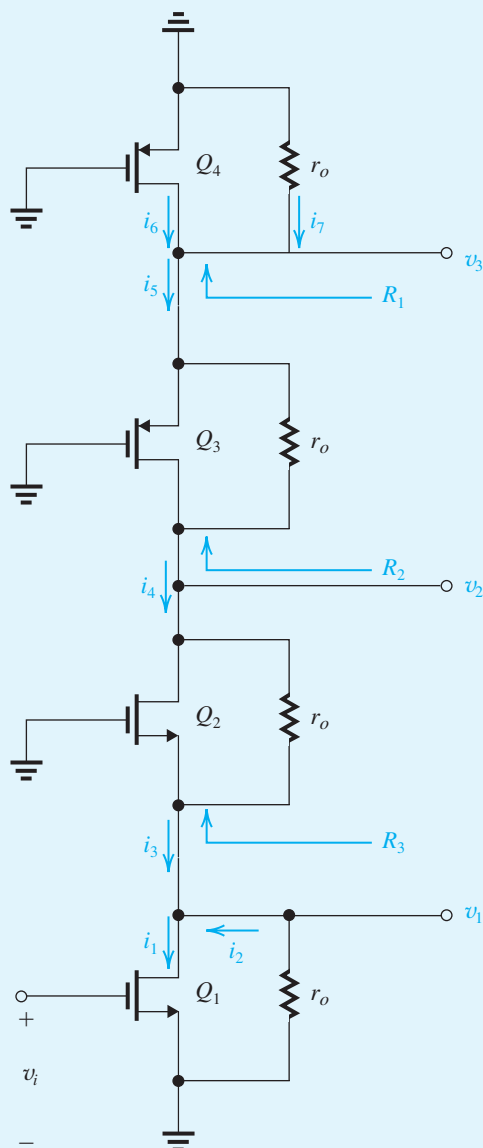


Figure P8.73

amplifier circuit. Knowledge of this signal distribution is very useful in designing the circuit so as to allow for the required signal swings. Figure P8.73 shows a CMOS cascode amplifier with all dc voltages replaced with signal grounds. As well, we have explicitly shown the resistance r_o of each of the four transistors. For simplicity, we are assuming that the four transistors have the same g_m and r_o . The amplifier is fed with a signal v_i .

- Determine R_1 , R_2 , and R_3 . Assume $g_m r_o \gg 1$.
- Determine i_1 , i_2 , i_3 , i_4 , i_5 , i_6 , and i_7 , all in terms of v_i . (Hint: Use the current-divider rule at the drain of Q_1 .)
- Determine v_1 , v_2 , and v_3 , all in terms of v_i .
- If v_i is a 5-mV peak sine wave and $g_m r_o = 20$, sketch and clearly label the waveforms of v_1 , v_2 , and v_3 .

D 8.74 Design the double-cascode current source shown in Fig. P8.74 to provide $I = 0.2 \text{ mA}$ and the largest possible signal swing at the output; that is, design for the minimum allowable voltage across each transistor. The $0.13\text{-}\mu\text{m}$ CMOS fabrication process available has $V_{tp} = -0.4 \text{ V}$, $V_A' = -6 \text{ V}/\mu\text{m}$, and $\mu_p C_{ox} = 100 \mu\text{A}/\text{V}^2$. Use devices with $L = 0.4 \mu\text{m}$, and operate at $|V_{ov}| = 0.2 \text{ V}$. Specify V_{G1} , V_{G2} , V_{G3} , and the W/L ratios of the transistors. What is the value of R_o achieved?

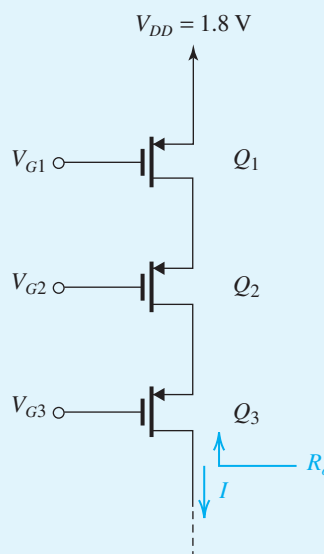


Figure P8.74

***8.75** Figure P8.75 shows a folded-cascode CMOS amplifier utilizing a simple current source Q_2 , supplying a current $2I$, and a cascoded current source (Q_4 , Q_5) supplying a current I . Assume, for simplicity, that all transistors have equal parameters g_m and r_o .

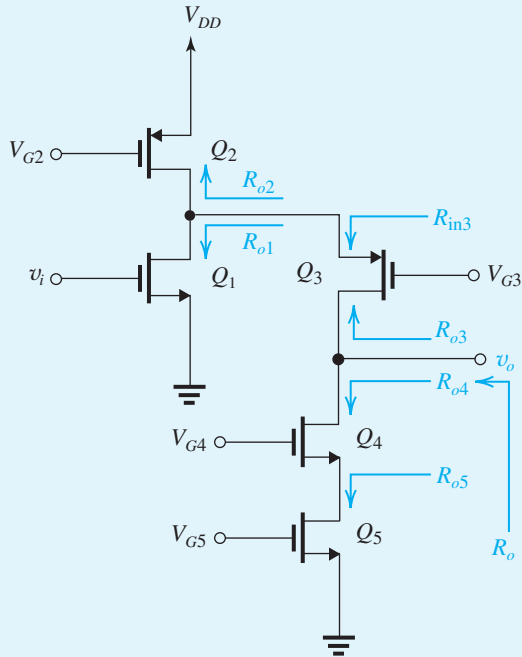


Figure P8.75

- Give approximate expressions for all the resistances indicated.
- Find the amplifier output resistance R_o .
- Show that the short-circuit transconductance G_m is approximately equal to g_{m1} . Note that the short-circuit transconductance is determined by short-circuiting v_o to ground and finding the current that flows through the short circuit, $G_m v_i$.

- Find the overall voltage gain v_o/v_i and evaluate its value for the case $g_{m1} = 2 \text{ mA/V}$ and $A_0 = 30$.

8.76 A cascode current source formed of two *pnp* transistors for which $\beta = 50$ and $V_A = 5 \text{ V}$ supplies a current of 0.2 mA . What is the output resistance?

8.77 Use Eq. (8.88) to show that for a BJT cascode current source utilizing identical *pnp* transistors and supplying a current I ,

$$IR_o = \frac{|V_A|}{(V_T/|V_A|) + (1/\beta)}$$

Evaluate the figure-of-merit IR_o for the case $|V_A| = 5 \text{ V}$ and $\beta = 50$. Now find R_o for the cases of $I = 0.1, 0.5$, and 1.0 mA .

8.78 Consider the BJT cascode amplifier of Fig. 8.38 for the case all transistors have equal β and r_o . Show that the voltage gain A_v can be expressed in the form

$$A_v = -\frac{1}{2} \frac{|V_A|/V_T}{(V_T/|V_A|) + (1/\beta)}$$

Evaluate A_v for the case $|V_A| = 5 \text{ V}$ and $\beta = 50$. Note that except for the fact that β depends on I as a second-order effect, the gain is independent of the bias current I !

8.79 A bipolar cascode amplifier has a current-source load with an output resistance βr_o . Let $\beta = 50$, $|V_A| = 100 \text{ V}$, and $I = 0.2 \text{ mA}$. Find the voltage gain A_v .

D *8.80 Figure P8.80 shows four possible realizations of the folded cascode amplifier. Assume that the BJTs have $\beta = 100$ and that both the BJTs and the MOSFETs have $|V_A| = 5 \text{ V}$.

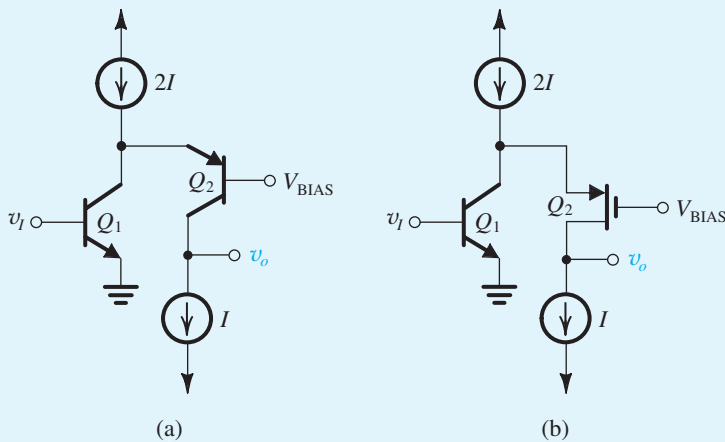


Figure P8.80

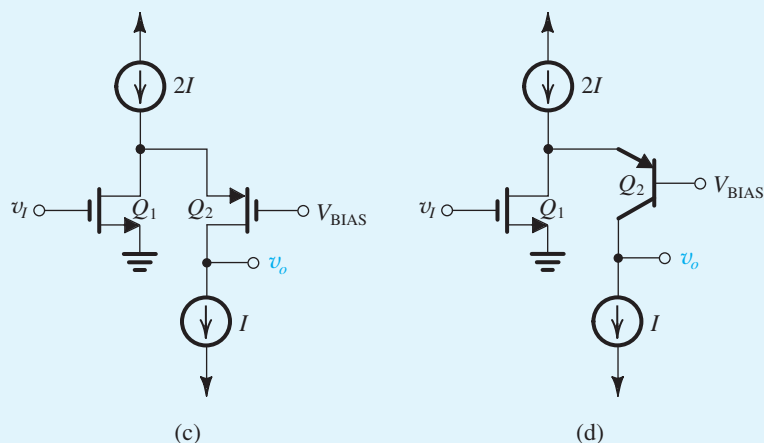


Figure P8.80 continued

Let $I = 100 \mu\text{A}$, and assume that the MOSFETs are operating at $|V_{OV}| = 0.2 \text{ V}$. Assume the current sources are ideal. For each circuit determine R_{in} , R_o , and A_{vo} . Comment on your results.

8.81 In this problem, we will explore the difference between using a BJT as cascode device and a MOSFET as cascode device. Refer to Fig. P8.81. Given the following data, calculate G_m , R_o , and A_{vo} for the circuits (a) and (b):

$$I = 100 \mu\text{A}, \beta = 125, \mu_n C_{ox} = 400 \mu\text{A/V}^2, W/L = 25, V_A = 1.8 \text{ V}$$

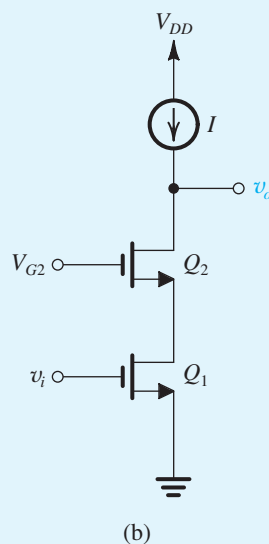
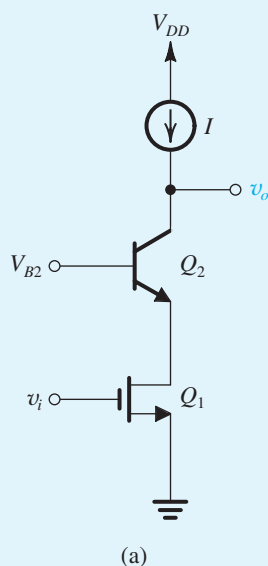


Figure P8.81 continued

Section 8.6: Current-Mirror Circuits with Improved Performance

SIM 8.82 In a particular cascoded current mirror, such as that shown in Fig. 8.39, all transistors have $V_t = 0.6 \text{ V}$, $\mu_n C_{ox} = 160 \mu\text{A/V}^2$, $L = 1 \mu\text{m}$, and $V_A = 10 \text{ V}$. Width $W_1 = W_4 = 4 \mu\text{m}$, and $W_2 = W_3 = 40 \mu\text{m}$. The reference current I_{REF} is $20 \mu\text{A}$. What output current results? What are the voltages at the gates of Q_2 and Q_3 ? What is the lowest voltage at the output for which current-source operation is possible? What are the values of g_m and

Figure P8.81

r_o of Q_2 and Q_3 ? What is the output resistance of the mirror?

8.83 Find the output resistance of the double-cascode current mirror of Fig. P8.83.

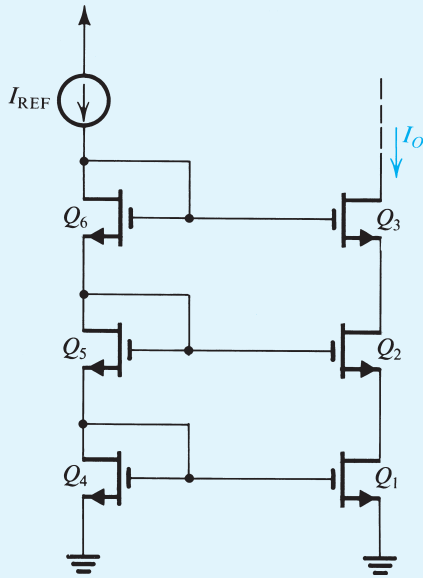


Figure P8.83

8.84 Consider the Wilson current-mirror circuit of Fig. 8.40 when supplied with a reference current I_{REF} of 1 mA. What is the change in I_O corresponding to a change of +10 V in the voltage at the collector of Q_3 ? Give both the absolute value and the percentage change. Let $\beta = 100$ and $V_A = 100$ V.

D 8.85 (a) The circuit in Fig. P8.85 is a modified version of the Wilson current mirror. Here the output transistor is “split” into two matched transistors, Q_3 and Q_4 . Find I_{O1} and I_{O2} in terms of I_{REF} . Assume all transistors to be matched with current gain β .
(b) Use this idea to design a circuit that generates currents of 0.1 mA, 0.2 mA, and 0.4 mA, using a reference current of 0.7 mA. What are the actual values of the currents generated for $\beta = 50$?

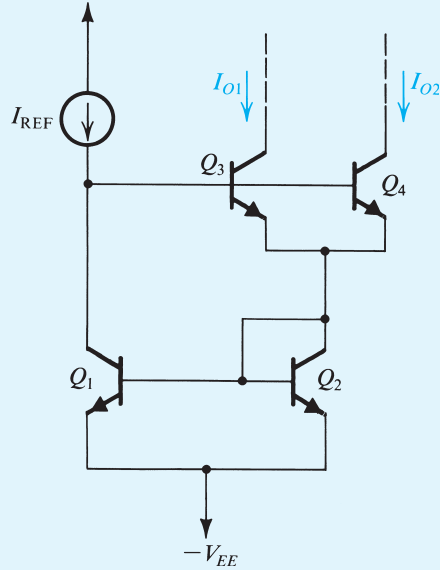


Figure P8.85

D 8.86 Use the *pnp* version of the Wilson current mirror to design a 0.1-mA current source. The current source is required to operate with the voltage at its output terminal as low as -2.5 V. If the power supplies available are ± 2.5 V, what is the highest voltage possible at the output terminal?

***8.87** For the Wilson current mirror of Fig. 8.40, show that the incremental input resistance seen by I_{REF} is approximately $2V_T/I_{REF}$. (Neglect the Early effect in this derivation and assume a signal ground at the output.) Evaluate R_{in} for $I_{REF} = 0.2$ mA.

***8.88** Consider the Wilson MOS mirror of Fig. 8.41(a) for the case of all transistors identical, with $W/L = 10$, $\mu_n C_{ox} = 400 \mu\text{A/V}^2$, $V_{tn} = 0.5$ V, and $V_A = 18$ V. The mirror is fed with $I_{REF} = 180 \mu\text{A}$.

- Obtain an estimate of V_{OV} and V_{GS} at which the three transistors are operating, by neglecting the Early effect.
- Noting that Q_1 and Q_2 are operating at different V_{DS} , obtain an approximate value for the difference in their currents and hence determine I_O .
- To eliminate the systematic error between I_O and I_{REF} caused by the difference in V_{DS} between Q_1 and Q_2 , a diode-connected transistor Q_4 can be added to the circuit